Quiz 2 - DSC 10, Spring 2025

Full Name:

PID:

Quiz Time: ◯ 3PM ◯ 3:30PM ◯ 4PM ◯ 4:30PM

Instructions:

- This quiz consists of 5 questions. You have a total of 20 minutes to complete it.
- Please write **clearly** in the provided answer boxes; we will not grade work that appears elsewhere. Completely fill in bubbles and square boxes; if we cannot tell which option(s) you selected, you may lose points.

  ◯ A bubble means that you should only **select one choice**.

  ☐ A square box means you should **select all that apply**.
- If your answer is a string, make sure to put it in quotes. If your answer is a float, make sure to include a decimal point.
- You may use one handwritten sheet of notes. No calculators and no computers.

By signing below, you are agreeing that you will behave honestly and fairly during and after this quiz.

Signature:

Version A

Please do not open your quiz until instructed to do so.

The DataFrame `ghibli` contains information about all 25 films produced by Studio Ghibli, the acclaimed Japanese animation studio. For each film, we have its `"Name"`, `"Director"`, `"Budget"` and `"Revenue"` (both in millions of dollars), primary `"Genre"`, and `"Runtime"` in minutes. `ghibli` has 25 rows. The first two are shown below.

	Name	Director	Budget	Revenue	Genre	Runtime
0	Spirited Away (2001)	Hayao Miyazaki	19.0	274.9	Animation	125
1	Howl's Moving Castle (2004)	Hayao Miyazaki	24.0	236.0	Fantasy	119

Question 1

Fill in the blanks below to add a new column to `ghibli` called `"TimeString"`, containing the runtime of each film as a string in hours and minutes. For example, *Spirited Away* is 125 minutes long, so its value in the `"TimeString"` column would be `"2 hours, 5 minutes"`.

```
def change_time(mins):
    hours = ___(a)___
    minutes = ___(b)___
    return str(hours) + " hours, " + str(minutes) + " minutes"

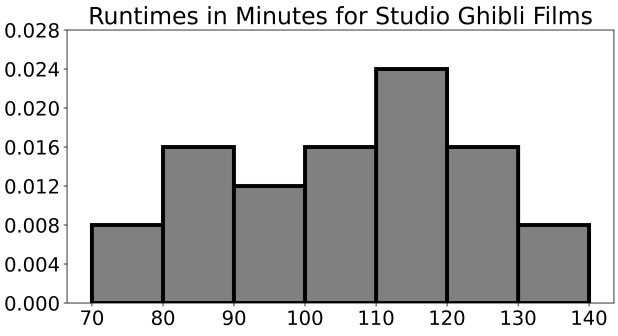
ghibli = ghibli.assign(TimeString = ___(c)___)
```

(a): (b):

(c):

Question 2

Below is a density histogram showing the distribution of `"Runtime"`, in minutes, for all 25 films in `ghibli`.



a) How many Studio Ghibli films are at least 90 minutes long?

b) Imagine we changed our histogram to have just one bin, from 70 to 140 minutes. What would the height of this bin be? Give your answer as an exact fraction or decimal.

PID: _____

Question 3

Using `ghibli`, write one line of code that outputs a DataFrame describing how many films of each genre each director has directed.

Question 4

Kate loves Studio Ghibli and she has been recording her viewings of their films in the `kate` DataFrame, which is shown in full below. Note that `kate` has repeated rows since Kate watched some films twice.

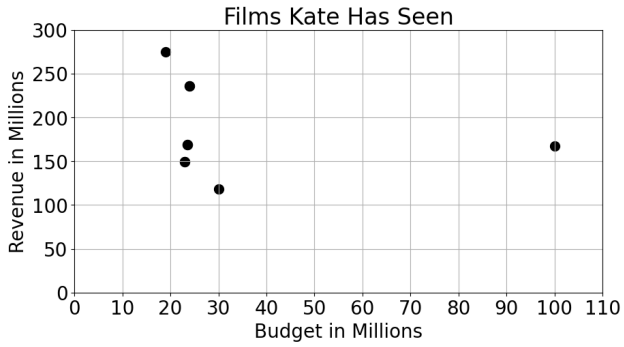
In `ghibli`, there are 11 films directed by Hayao Miyazaki and only 2 directed by Hiromasa Yonebayashi.

How many rows are in the DataFrame resulting from `ghibli.merge(kate, on="Director")`?

	Name	Director
0	Spirited Away (2001)	Hayao Miyazaki
1	The Secret World of Arrietty (2010)	Hiromasa Yonebayashi
2	The Wind Rises (2013)	Hayao Miyazaki
3	The Secret World of Arrietty (2010)	Hiromasa Yonebayashi
4	Princess Mononoke (1997)	Hayao Miyazaki
5	Howl's Moving Castle (2004)	Hayao Miyazaki
6	The Boy and the Heron (2023)	Hayao Miyazaki
7	Howl's Moving Castle (2004)	Hayao Miyazaki
8	Princess Mononoke (1997)	Hayao Miyazaki

Question 5

The scatter plot below shows the budget and revenue (in millions of dollars) for each of the 6 distinct films in `kate`. Answer each question below, or write "not enough info" if you don't have enough information to answer.



a) How many of these films had less than 200 million dollars in revenue?

b) How many of these films earned a revenue more than five times their budget?

Reminders:

- **Write your PID** on the front page and on the top right corner of page 3.
- If your answer is a string, make sure to put it in **quotes**.
- If your answer is a float, make sure to include a **decimal point**.
- Stay in your seat until the quiz is over. No leaving early.